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Shapley's Value as a Resource Optimization Strategy for Digital Radio Transmission over IBOC FM

Mónica Rico Martínez , Juan Carlos Vesga Ferreira * , Joel Carroll Vargas, María Consuelo Rodríguez ,
Andres Alejandro Diaz Toro  and William Alexander Cuevas Carrero

Telecommunications Engineering Program, School of Basic Sciences, Technology and Engineering (ECBTI), Universidad Nacional Abierta y a Distancia, Bogotá 111321, Colombia; monica.rico@unad.edu.co (M.R.M.); joel.carroll@unad.edu.co (J.C.V.); mariac.rodriguez@unad.edu.co (M.C.R.); andres.diaz@unad.edu.co (A.A.D.T.); william.cuevas@unad.edu.co (W.A.C.C.)

* Correspondence: juan.vesga@unad.edu.co; Tel.: +57-3212759948

Abstract: The hybrid in-band on-channel (IBOC) transmission system, which serves as a digital audio radio scheme, facilitates the simultaneous transmission of analog FM and digital audio. In IBOC systems, broadcasters transmit signals within allocated channel bands, posing a challenge in bandwidth allocation for digital audio transmission. To address this, cooperative game theory, supported by the bankruptcy game and Shapley's value, is proposed as a strategy to optimize bandwidth allocation at each node or station. This approach considers service demand, the number of stations, and radio channel conditions in the FM band. The paper presents a scenario under saturated traffic conditions to evaluate the degree of optimization achievable using the Shapley value under clearly defined traffic and channel conditions. The results indicate that cooperative game theory, with the support of the Shapley value, offers an excellent alternative for optimizing bandwidth in digital broadcasting over IBOC in FM, with potential applicability in other digital radio broadcasting systems.

Keywords: IBOC; FM; Shapley; optimization; game theory; bandwidth allocation



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1. Introduction

Due to the emergence of new high-fidelity multimedia services accompanied by optimal use of the electromagnetic spectrum, there has been a need to initiate an evolutionary migration process from traditional analog to digital broadcasting. In recent years, several digital audio transmission standards have emerged, each leveraging various technological approaches. One of the most widely accepted standards, due to the evolutionary transition from analog AM and FM radio to digital radio, is IBOC technology (NRSC-5). It has the ability to offer sound quality similar to that of a CD, accompanied by a variety of multimedia services, such as data, images, and even video [1].

Carrying out a migration process from analog to digital radio in an evolutionary manner can be considered a complex process due to the number of challenges that must be resolved. These challenges include finding adequate coexistence between analog and digital radio within the bandwidth established for each AM or FM radio channel, reducing interference levels during analog signal reception, and evaluating appropriate power levels for the transmission of the two types of signals, among other aspects [2]. Additionally, an optimal digital radio system must offer efficiency and spectral flexibility, with the ability to provide high transmission speed rates without monopolizing continuous bandwidth within a dynamic broadcasting context [3].

The in-band on-channel (IBOC) system is designed to operate in two specific modes: hybrid mode and fully digital mode. In the hybrid mode, the analog and digital components are transmitted simultaneously, with both signals using the channel assigned to the analog

signal under the traditional radio model. In the fully digital mode, the transmitted signal utilizes the entire bandwidth assigned to the station, and its components are purely digital.

IBOC adopts a hierarchical modulation scheme to further enrich Broadband Digital Radio (BDR) services. Hierarchical transmission provides layered services to mobile users within different coverage areas through the same radio channel. For example, a basic layer signal is used to transmit essential information (normal quality audio/video), while the secondary layer, added on top of the basic layer, serves as a transport medium for complementary multimedia services [4]. Therefore, hierarchical streaming offers an attractive solution for terminals with different screen sizes and video display resolutions. However, the advantages of hierarchical streaming come with a cost. Most notably, the secondary layer reduces the effective transmission power of the basic layer, leading to additional performance loss in the reception of the basic layer [5].

In conclusion, while there are several digital radio systems, each with specific advantages, all offer significant benefits compared to analog radio. Digital radio, utilizing perceptual audio coding, can provide better sound quality than AM or FM radio. Based on the Orthogonal Frequency Division Multiplexing (OFDM) multi-carrier transmission technique, digital radio can overcome mobile reception distortion caused by multipath propagation and other interference inherent in analog radios. Unlike analog radio, digital radio allows for the inclusion of additional services such as text and/or multimedia information transmission. In fact, nearly anything that can be digitized can be transmitted over a digital radio platform [6]. A crucial aspect is that these systems use MPEG-2 audio coding, facilitating the transmission of up to five programs with near CD quality in a set. Additionally, by synergistically combining Low-Density Parity Check (LDPC) codes, band aggregation, frequency hopping techniques, and an efficient hierarchical modulation scheme, IBOC offers performance and service advantages in various application scenarios, accompanied by a wide range of transmission rates (up to 2.53 Mbps), and the ability to provide high-quality radio and rich multimedia services [7].

Given the above, it is proposed to employ cooperative game theory, supported by the use of the Shapley value, as a strategy for bandwidth allocation to each of the nodes or stations that are part of the broadcasting network over IBOC FM. This aims to improve performance by allowing multiple nodes to transmit simultaneously under clearly defined distribution policies [8].

The article follows a clear and orderly structure, describing in Section 2 the fundamentals of digital broadcasting under IBOC FM technology, as well as the elements that are part of the Shapley value and the Bankruptcy game, as a strategy for equitable resource optimization supported by the use of Cooperative Game Theory. Subsequently, in Section 3, it proposes a scenario similar to a real context where each station defines bandwidth requirements, with the particularity that the total bandwidth requested exceeds the available bandwidth (state of saturation). This will be analyzed using the Shapley value and resource optimization evaluation criteria, with the results compared to the optimization technique of linear programming to evaluate the quality of the optimization process carried out by the Shapley value. Finally, in Sections 4 and 5, the discussion of the results obtained and the conclusions reached by the manuscript authors on the use of the proposed techniques applied to the specific scenario in IBOC FM are presented, highlighting the improvements achieved in bandwidth allocation and system efficiency. The discussion will focus on interpreting and contextualizing these results in relation to the existing literature, exploring theoretical and practical implications, advantages, and disadvantages, among other aspects, highlighting the relevant aspects obtained and their importance in the field of digital broadcasting, as well as possible future research.

2. Materials and Methods

2.1. Game Theory and Shapley Value

Game theory, conceptualized by John Von Neumann in 1928, offers a mathematical framework for assessing individual decisions within competitive contexts of gain or loss

compared to decisions made by other competitors. This competitive scenario is referred to as a “Game”, and the individuals who are part of this scenario are called “Players”. Game theory provides three models for representing real scenarios: extensive, strategic, and coalition. While the first two are applicable to non-cooperative games, emphasizing individual player interests, the coalition model pertains to cooperative games. In cooperative games, players collaborate to achieve common goals, increasing the likelihood of higher gains than individual actions. Cooperative game analysis avoids the need to scrutinize player strategies, focusing instead on coalition utility and payment vectors [9].

In cooperative game theory, a common question arises concerning the fair distribution of a good’s net value among players, particularly in cases of insufficient resources. This scenario, known as the “bankruptcy” problem, is typically analyzed either as a transferable utility game or through axiomatic methods [10,11]. This paper adopts the transferable utility approach.

Definition 1 ([12]). A Transferable Utility (TU) cooperative game is a pair (N, v) , where N is the set of players $\{1, 2, \dots, n\}$ and $v: 2N \rightarrow \mathbb{R}$ is the characteristic function, with $v(\emptyset) = 0$. Each coalition $S \subset N$ associates with $v(S)$, representing the assured payoff for its players, irrespective of others’ actions.

Definition 2 ([13]). A bankruptcy game, represented as (N, d, E) , comprises creditors (N), claims vector (d), and net value (E). A cooperative game (N, v) is defined from each bankruptcy problem. The value of coalition S is the property not claimed by non- S members, calculated as follows:

$$v(S) = \max\{0, E - d(N \setminus S)\} \quad \forall S \subset N \quad (1)$$

In TU cooperative game theory, a major challenge lies in the equitable allocation of the total gain among players. The Shapley value, introduced by Shapley in 1953, tackles this issue by accounting for each player’s marginal contributions to all potential coalitions, satisfying established axioms [14,15]. The expression corresponding to the Shapley value is as follows:

$$\varphi_i(v) = \sum_{S \subseteq N: i \in S} \frac{(s-1)!(n-s)!}{n!} [v(S) - v(S - \{i\})] \quad \text{where } n = |N| \text{ y } s = |S| \quad (2)$$

The Shapley value guarantees efficient, symmetric, passive, and additive resource allocations. O’Neill (1982) demonstrated the correspondence between the recursive partitioning process in a bankruptcy problem and Shapley’s value. For the solution to be adequate, it is necessary for the vector of payoffs to comply with the efficiency principle, as outlined in [16]:

$$\sum_{i \in N} \varphi_i(v) = v(N) \quad (3)$$

$$\varphi_i(v) \geq (v\{i\}) \quad \forall i \in N \quad (4)$$

In the context of equitable bandwidth distribution in the digital radio spectrum for IBOC FM, represented as a TU game (N, v) , the Shapley value emerges as a strategy for fair allocation among nodes under saturation conditions. This scenario involves N nodes as players, where λ_i represents the bandwidth demand of each node i (d_i) and $E = BW_T$.

2.2. Digital Radio on IBOC

Radio serves as a public service with a profound impact on the social development of a country’s inhabitants, aiming to ensure the right to information, recreation, free expression, and personal growth. To harness the benefits of digitalization, countries worldwide have initiated efforts to accelerate digital terrestrial migration by formulating national plans to establish digital broadcasting infrastructure and phase out analog services. Leading countries such as Norway, Denmark, Switzerland, the USA, and Japan have spearheaded various radio standards as needed, with some announcing digital transition

dates. Progress in this transition has been steady, with countries like Germany and the Netherlands embracing digital technology, while others like Spain, Sweden, and Ireland adopting a wait-and-see approach. Successful implementations of digital radio standards have occurred in the US and Japan. One of the significant challenges in choosing a digital radio standard is making the right decision considering all technical aspects to meet each country's needs and demands [17].

Currently, digital radio penetration worldwide is predominant in North America, where the IBOC system is chosen, and in Europe, where the DAB standard is adopted. Latin America lacks a precedent, but countries like Argentina and Chile have shown interest in radio digitalization [18]. Consequently, numerous countries are in the implementation process or interested in transitioning from analog to digital broadcasting. It is crucial to understand the characteristics of these technologies to select the most suitable one based on existing broadcasting conditions, user needs, and country regulations.

In most countries, implementing a digital broadcasting system requires resolving technical and regulatory aspects. Firstly, regarding frequencies for the chosen digital broadcasting system, determining working frequencies is necessary to enable the simultaneous use of analog and digital receivers. This situation favors IBOC technology over others such as DAB, which necessitates a broader scale change [19]. Secondly, in the regulatory domain, mechanisms for frequency allocation, license granting, and participation of new stakeholders need to be defined. Given the experiences in the United States and recommendations from the FCC, IBOC may be the likely technology for digital broadcasting implementation in countries like Colombia. Hence, a detailed description of the communication protocols within this standard for audio coding and transmission in a digital context is presented below.

2.3. IBOC FM Spectrum for Hybrid Mode

In the hybrid model, the digital signal is transmitted in the primary main sidebands on both sides of the FM signal. In this model, each sideband of the IBOC FM system comprises 10 frequency divisions allocated between carriers 356 to 545 and between -356 to -545 . Carriers 546 and -546 serve as reference carriers. Furthermore, the level of the digital subcarriers describes a total power of 23 dB below the nominal power of the analog FM carrier [2]. Figure 1 illustrates the carrier distribution for the hybrid mode.

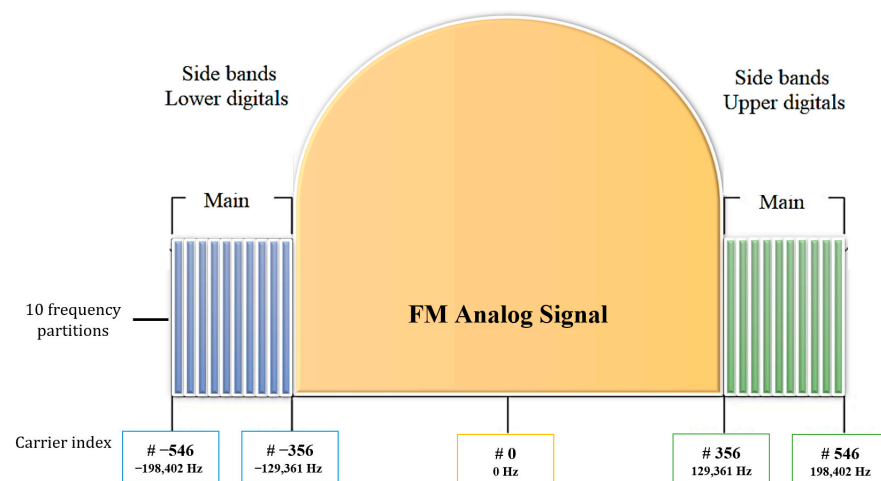


Figure 1. IBOC FM hybrid waveform spectrum.

2.4. Spectrum for the Extended Hybrid Mode

In the extended hybrid mode, additional sidebands are introduced alongside the primary sidebands of the hybrid mode, resulting in an expansion of the hybrid sidebands toward the central area where the analog FM component is situated. This modification enhances the digital capacity of the system. However, it impacts analog coverage as the

analog bandwidth is reduced. Depending on the service mode, one, two, or four frequency divisions are added at the inner edge of the primary main sideband, thereby reducing the bandwidth allocated to the analog FM signal [20]. Moreover, in addition to the subcarrier structure described in the hybrid mode, the extended sidebands encompass subcarriers ranging from -356 to -280 and from 280 to 356 . The amplitude of the subcarriers within the extended sidebands matches that of the main primary subcarriers. Figure 2 depicts the spectrum corresponding to the IBOC FM extended hybrid model [5].

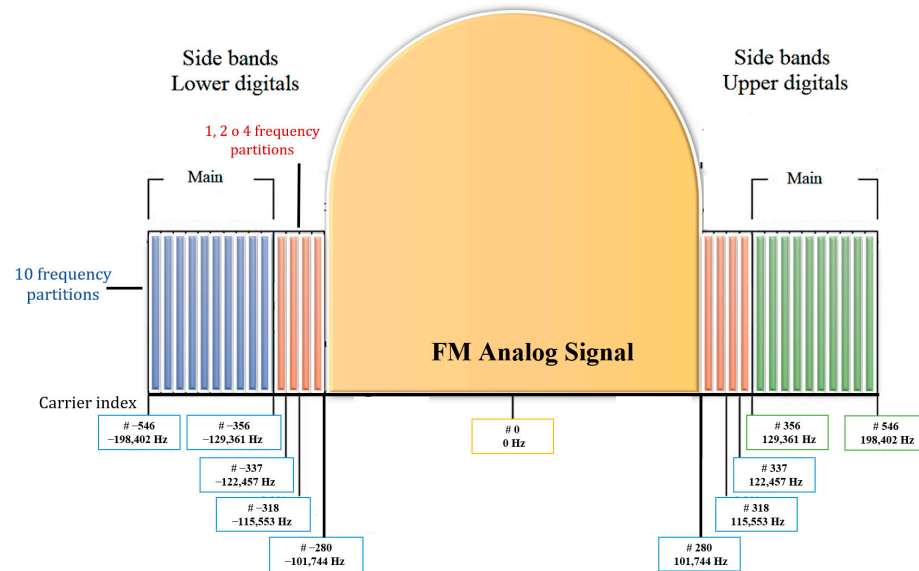


Figure 2. Spectrum corresponding to the IBOC FM extended hybrid model.

2.5. Spectrum for All-Digital Mode

In the all-digital mode, the spectrum is constructed by completely eliminating the analog signal, thereby allowing the entire bandwidth to be utilized for secondary sidebands. This scheme facilitates an enhanced mode of operation, wherein broadcasters transition from the hybrid system to the all-digital system, thereby achieving superior quality levels [21]. Figure 3 illustrates the carrier structure for the all-digital mode, with each of the secondary sidebands comprising 10 main frequency divisions and 4 extended frequency divisions.

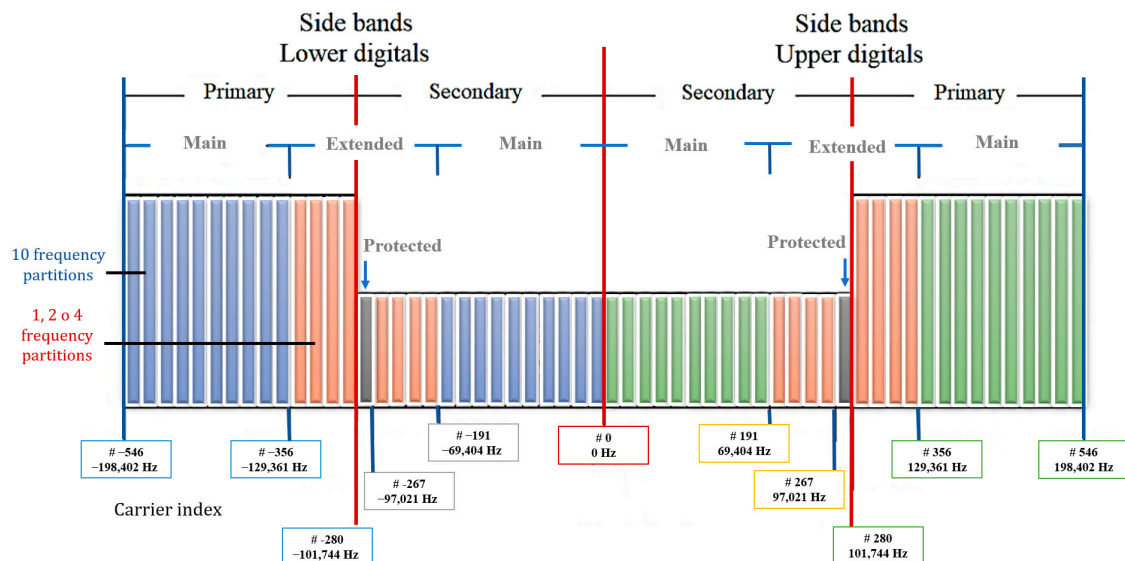


Figure 3. Spectrum of the fully digital IBOC FM waveform.

Two reference carriers, numbered -279 and 279 , and twelve protected OFDM subcarriers are transmitted in an area of the spectrum least likely to be interfered with by analog and digital interference. Each secondary sideband comprises carriers 1 to 190 and -1 to -190 . The extended secondary sidebands encompass subcarriers 191 to 266 and -191 to -266 . The protected sidebands consist of subcarriers -267 to -278 and 267 to 278 . The average power of the secondary subcarriers will be between 5 and 20 dB below the primary subcarriers [1].

In the IBOC FM system, different service modes associated with the operating modes and logical channels are defined. For the hybrid mode, the operating modes range from MP1 to MP7. In turn, for the all-digital operating mode, service modes ranging from MS1 to MS4 are available. IBOC has four main logic channels (P1, P2, P3, and PIDS) and six secondary logic channels (S1, S2, S3, S4, S5, and SIDS). A logical channel is a signal path carrying data frames with a given quality of service. Channels P1, P2, P3, and P4 configure the different primary audio services, and the PIDS channel provides the primary data service (IDS). Channels S1, S2, S3, S4, and S5 are used exclusively in the all-digital system for the transmission of data or ambient sound (supplementary audio). The SIDS provides the secondary data service [22].

The system control channel (SCCH) carries the control and status information related to the operating mode and configuration parameters. Additionally, the OFDM subcarriers are arranged in groups called frequency divisions. Each frequency division consists of 18 data subcarriers and one reference subcarrier, with a subcarrier spacing of 363.4 Hz. Table 1 describes the most important parameters of the IBOC FM system [3]. Tables 2 and 3 depict the characterization of the primary and secondary logical channels, respectively, according to the established service mode for transmission.

Table 1. IBOC FM system parameters.

Parameter	Symbol	Units	Exact Value	Approximate Value
OFDM subcarrier spacing	Δf	Hz	$\frac{1488375}{4096}$	363.4
Predefined cyclic width	α	-	$\frac{7}{128}$	5.469×10^{-2}
OFDM symbol duration	T_s	s	$\frac{1+\alpha}{\Delta f}$	2.902×10^{-3}
OFDM symbol rate	R_s	Hz	$\frac{1}{T_s}$	344.5
L1 frame duration	T_f	s	$512 \cdot T_s$	1.486
L1 frame speed	R_f	Hz	$\frac{1}{T_f}$	6.729×10^{-1}
L1 block duration	T_b	s	$32 \cdot T_s$	9.288×10^{-2}
L1 block speed	R_b	Hz	$\frac{1}{T_b}$	10.77
Duration of even block L1	T_p	s	$64 \cdot T_s$	1.858×10^{-1}
Torque block speed L1	R_p	Hz	$\frac{1}{T_p}$	5.383

Table 2. Characterization of primary logic channels according to service mode (IBOC FM).

Service Mode	Logical Channel	Transfer			L1 Latency [s]	Relative Robustness
		Frame Size [bits]	Frame Speed [Hz]	Frame Module		
MP1	P1	146.176	R_f	1	T_f	2
	PIDS	80	R_b	16	T_b	3
MP2	P1	146.176	R_f	1	T_f	2
	P3	2.304	R_p	8	$2T_f$	3
	PIDS	80	R_b	16	T_b	3

Table 2. Cont.

Service Mode	Logical Channel	Transfer			L1 Latency [s]	Relative Robustness
		Frame Size [bits]	Frame Speed [Hz]	Frame Module		
MP3	P1	146.176	R_f	1	T_f	2
	P3	4.608	R_p	8	$2T_f$	3
	PIDS	80	R_b	16	T_b	3
MP4	P1	176.176	R_f	1	T_f	2
	P3	9.216	R_p	8	$2T_f$	3
	P4	9.216	R_p	8	$2T_f$	3
	PIDS	80	R_b	16	T_b	3
MP5	P1	4.608	R_p	8	$T_p + T_{dd}$	1
	P2	109.312	R_f	1	T_f	2
	P3	4.608	R_p	8	$2T_f$	3
	PIDS	80	R_b	16	T_b	3
MP6	P1	9.216	R_p	8	$T_p + T_{dd}$	1
	P2	72.448	R_f	1	T_f	2
	PIDS	80	R_b	16	T_b	3

Table 3. Characterization of the secondary logic channels according to the service mode (IBOC FM).

Service Mode	Logical Channel	Transfer			L1 Latency [s]	Relative Robustness
		Frame Size [bits]	Frame Speed [Hz]	Frame Module		
MS1	S4	18.272	R_p	8	T_p	7
	S5	512	R_b	16	T_b	6
	SIDS	80	R_b	16	T_b	8
MS2	S1	4.608	R_p	8	$T_p + T_{dd}$	5
	S2	109.312	R_f	1	T_f	9
	S3	4.608	R_p	8	T_p	11
	S5	512	R_b	16	T_b	6
	SIDS	80	R_b	16	T_b	10
	S1	9.216	R_p	8	$T_p + T_{dd}$	5
MS3	S2	72.448	R_f	1	T_f	9
	S5	512	R_b	16	T_b	6
	SIDS	80	R_b	16	T_b	10
	S1	4.608	R_p	8	T_p	11
MS4	S2	146.176	R_f	1	T_f	9
	S3	4.608	R_p	8	T_p	11
	S5	512	R_b	16	T_b	6
	SIDS	80	R_b	16	T_b	10
	S1	4.608	R_p	8	T_p	11

2.6. Audio Source Encoding and Compression in IBOC

The human voice (analog), to be transmitted efficiently through any medium in a digital format, requires a transformation process known as “digitization”, which is performed by a device called CoDec (Codec/Decoder), where each transmission mechanism may have several alternatives of Codecs, as appropriate. Considering that, for the transport of voice information, it is necessary to assemble the information in the form of packets or PDUs, the required bandwidth will depend on the overhead generated by these packets [23].

IBOC uses a Perceptual Audio Coder (PAC) encoding algorithm from Lucent Technologies, which is a coding method that uses advanced signal processing techniques and the use of psychoacoustic models, through which it is possible to obtain a high compression of the source signal. It uses a sampling frequency of 44.1 kHz, 16 bits of resolution, and, by means of a bank of special filters, it obtains a complete description of the audio signal whose processing facilitates an auditory perception analysis for its quantification and subsequent coding, optimizing the transmitted information to the extreme [24]. One of the main benefits of PAC is that it allows audio compression at different bit rates, even as low as 6 kbps. When the channel quality is 6 to 8 kbit/s, the result is similar to an AM signal. For a channel capacity between 16 and 24 kbit/s, the result is similar to an FM signal, and when a capacity of 32 or 64 kbps is available, the audio quality is considered very similar to what can be obtained on a compact disc. In the particular case of IBOC FM, it is possible to perform transmission processes of high-quality audio channels (comparable to compact disc) with transmission speeds of only 96 kb/s. In turn, the IBOC FM system (standardized as NRSC-5) does not propose a specific source encoder, but makes use of nominal and minimum rates of the source encoders for each of the operating modes [25].

IBOC utilizes a Perceptual Audio Coder (PAC) encoding algorithm developed by Lucent Technologies. This coding method employs advanced signal processing techniques and psychoacoustic models to achieve high compression of the source signal, operating at a sampling frequency of 44.1 kHz with 16 bits of resolution and utilizing a bank of specialized filters. PAC provides a comprehensive description of the audio signal, facilitating auditory perception analysis for quantification and subsequent coding. This optimization of transmitted information allows for significant compression, even at bit rates as low as 6 kbps [24].

PAC offers several benefits, including the ability to compress audio at different bit rates. For channel quality ranging from 6 to 8 kbit/s, the result resembles an AM signal. In the case of channel capacities between 16 and 24 kbit/s, the quality is akin to an FM signal, while capacities of 32 or 64 kbps yield audio quality similar to that of a compact disc. Remarkably, in the context of IBOC FM, high-quality audio channels comparable to compact discs can be transmitted at speeds as low as 96 kb/s. It is noteworthy that the IBOC FM system (standardized as NRSC-5) does not propose a specific source encoder but instead relies on the nominal and minimum rates of the source encoders for each operating mode [25].

2.7. Estimation of the Transmission Rate Required by the IBOC FM System

To estimate the values corresponding to the calculation of the transmission rate, the following expression is utilized in accordance with the recommendations of the standard [NRSC-5B] and the values documented in Tables 1–3. The expression is as follows:

$$V_{tx}[\text{bps}] = \text{frame size}[\text{bits}] \cdot \text{frame speed}[\text{Hz}] \quad (5)$$

For example, if you wish to estimate the baud rate for IBOC FM in service mode MP1, and output through logic channel P1:

$$V_{tx}[\text{bps}] = 146176 \left[\frac{44100}{65536} \right] \approx 98.4[\text{kbps}] \quad (6)$$

Tables 4 and 5 display the required transmission rates of the primary and secondary logical channels respectively, based on the service mode.

Table 4. Transmission rates of primary logic channels.

Service Mode	Approximate Transfer Speed [kbps]					Waveform
	P1	P2	P3	P4	PIDS	
MP1	98	N/A	N/A	N/A	1	Hybrid
MP2	98	N/A	12	N/A	1	Extended hybrid
MP3	98	N/A	25	N/A	1	Extended hybrid
MP4	98	N/A	25	25	1	Extended hybrid
MP5	25	74	25	N/A	1	All-digital extended hybrid
MP6	50	49	N/A	N/A	1	All-digital extended hybrid

N/A means not applicable.

Table 5. Transmission rates of secondary logic channels.

Service Mode	Approximate Transfer Speed [kbps]						Waveform
	S1	S2	S3	S4	S5	SIDS	
MS1	0	0	0	98	6	1	Hybrid
MS2	25	74	25	0	6	1	Extended hybrid
MS3	50	49	0	0	6	1	Extended hybrid
MS4	25	98	25	0	6	1	Extended hybrid

3. Results

3.1. Description of the Proposed Scenario

To comprehend the utilization of the Shapley value as a strategy for resource optimization within an IBOC broadcasting scheme in FM, a scenario is outlined involving twelve (12) nodes or digital broadcasting stations conducting IBOC hybrid transmission processes over FM. These transmissions occur within an RF channel that, for this specific case, is assumed to provide a total bandwidth (BW) of 1150 kbps due to conditions affecting the channel's maximum performance.

Recorded in Table 6 is each class of traffic per node, where it can be observed that the total bandwidth required is 1387 kbps, exceeding the total bandwidth available in the RF channel. This indicates that the operation mode will be in a state of saturation. To calculate the value of BW, the expression provided by Equation (7) is used:

$$BW_T = \frac{1}{T_s} \sum_{k=1}^{N_{sp}} \log_2 \left[1 + \frac{SNR_k}{\Gamma} \right] \quad (7)$$

N_{sp} : Number of subcarriers.

T_s : Time of an OFDM symbol ($T_s = 2.902$ ms) for the particular case of IBOC FM.

SNR_k : Signal to Noise Ratio (SNR) present on the subcarrier k .

Γ : This is known as the SNR gap, which represents the loss in SNR incurred by using a specific discrete coding scheme. It is suggested in [26] that the value of Γ can be calculated for practical purposes using Equation (8):

$$\Gamma = -\frac{1}{1.6} \ln \left[\frac{BER_{obj}}{0.2} \right] \quad (8)$$

where the BER target (BER_{obj}) corresponds to the desired Bit Error Rate value to be sustained, which, in this particular case, is set to 10^{-6} . For the proposed scenario, we

consider $N = 12$ nodes comprising the RF system in a saturated state ($BW_T \leq \sum_{i=1}^N d_i$), with a total available bit rate $BW_T = E = 1150$ kbps.

Table 6. BW required at each node according to service mode.

Node i	Service Mode	Logic Channels BW Required [kbps].				
		P1	P2	P3	P4	PIDS
1	MP1	98				
2	MP2	98		12		
3	MP3	98		25		1
4	MP3	98		25		
5	MP5	25	74	25		1
6	MP1	98				
7	MP4	98		25	25	1
8	MP6	50	49			1
9	MP2	98		12		
10	MP5	25	74	25		1
11	MP5	25	74	25		1
12	MP6	50	49			1

As mentioned above, a bankrupt set will be considered consistent with the saturation state of the RF channel to calculate the transferable utility value for each of the coalitions. Algorithm 1 to perform the calculation of each of the transferable utility values is as follows:

Algorithm 1: CalculateCoalitions

Nj: Number of Players

M_Coalitions: Matrix of possible coalitions

V_Coalition: Transferable utility value for each coalition

1. Initialize Z as a vector containing numbers from 1 to Nj.
2. Set n_coal to 0.
3. For each coalition size i from 1 to Nj:
 - a. Increment n_coal by n choose i, where n is Nj.
4. Initialize M_Coalitions as a matrix of zeros with dimensions (n_coal $\times$ Nj).
5. Initialize c (counter) to 0.
6. For each coalition size i from 1 to Nj:
 - a. Generate all combinations S of Z taken i at a time.
 - b. For each combination S:
 - i. Increment the counter.
 - ii. Initialize Suma_d to 0.
 - iii. For each member k in S:
 - Suma_d += V(k)
 - M_Coalitions(c, k) = k
 - iv. Calculate Suma_dT as BW_T minus (Total_V minus Suma_d).
 - v. Set V_Coalition(c) to the maximum of 0 and Suma_dT.

End of Algorithm.

Table 7 presents the transferable utility values calculated for each of the possible coalitions in the saturation state, utilizing the bankruptcy game. The values $v(S)$ consider that the principle of rational individuality must be satisfied (N, v), and the principle of efficiency is $\sum_{i \in S} \varphi_i(v) \geq v(S) \forall S \subseteq N, \sum_{i \in N} \varphi_i(v) = v(N)$.

Table 7. Transferable utility value for each of the coalitions $v(S)$.

Coalition	Value [1×10^3]	Coalition	Value [1×10^3]
{1}	0.00	{7}	0.00
{2}	0.00	{8}	0.00
{3}	0.00	{9}	0.00
{4}	0.00	{10}	0.00
{5}	0.00	{11}	0.00
{6}	0.00	{12}	0.00
Grand coalition		{1,2,3,4,5,6,7,8,9,10,11,12}	1150

Therefore, the equations that describe each one of the imputations for the game (N, v) are as follows:

$$\begin{aligned}
 \varphi_1(v) &\geq 0 \\
 \varphi_2(v) &\geq 0 \\
 \varphi_3(v) &\geq 0 \\
 \varphi_4(v) &\geq 0 \\
 \varphi_5(v) &\geq 0 \\
 \varphi_6(v) &\geq 0 \\
 \varphi_7(v) &\geq 0 \\
 \varphi_8(v) &\geq 0 \\
 \varphi_9(v) &\geq 0 \\
 \varphi_{10}(v) &\geq 0 \\
 \varphi_{11}(v) &\geq 0 \\
 \varphi_{12}(v) &\geq 0
 \end{aligned} \tag{9}$$

The efficiency equation for the proposed scenario is as follows:

$$\begin{aligned}
 \varphi_1(v) + \varphi_2(v) + \varphi_3(v) + \varphi_4(v) + \varphi_5(v) + \varphi_6(v) + \varphi_7(v) + \varphi_8(v) + \\
 \varphi_9(v) + \varphi_{10}(v) + \varphi_{11}(v) + \varphi_{12}(v) = 1150 * 10^3
 \end{aligned} \tag{10}$$

Based on the above expressions, the core of the game would be as follows:

$$C(v) = \left\{ \varphi(v) \in \mathbb{R}^n \mid \sum_{i \in S} \varphi_i(v) \geq v(S) \ \forall S \subseteq N, \ \varphi_1(v) + \varphi_2(v) + \varphi_3(v) + \varphi_4(v) + \right. \\
 \left. + \varphi_5(v) + \varphi_6(v) + \varphi_7(v) + \varphi_8(v) + \varphi_9(v) + \varphi_{10}(v) + \varphi_{11}(v) + \varphi_{12}(v) = v(N) \right\} \tag{11}$$

The Algorithm 2 to calculate the Shapley value $\varphi_k(v) \ \forall k \in N$ is as follows:

Table 8 displays the result corresponding to the Shapley matrix in the saturation state for the proposed scenario. The last column presents the Shapley value calculated for each of the players. Additionally, it can be observed that $\sum_{i \in N} \varphi_i(v) = v(N) = BW_T$.

Figure 4 illustrates the values for each node corresponding to the requested bandwidth and the bandwidth allocated. It should be noted that the total bandwidth demanded by the RF network exceeds the total bandwidth available in the channel. Therefore, the values assigned to each node should be lower than the demanded value. Additionally, it can be observed that the bandwidth requested by each node exceeds the amount assigned by the optimization process, consistent with a scenario in a state of saturation.

Algorithm 2: ShapleyValueCalculation

Nj: Number of Players
 M_Coalitions: Matrix of possible coalitions
 V_Coalition: Transferable utility value for each coalition
 Mk_Coalitions: Matrix of coalitions associated with player k
 Vk_Coalition: TU value for each coalition associated with player k
 Sub_coal: Matrix of subcoalitions formed by deleting player k
 Vk_Sub_coal: TU value for each subcoalition
 M_Shapley: Shapley matrix
 B: Vector of Shapley coefficients (P(j))
 Weight: Vector of Shapley values for each player k

1. For each player k in [1, Nj]:
 - a. For each coalition size lg in [1, Nj]:
 - i. Initialize nMk to 0.
 - ii. For each coalition i in [1, n_coal]:
 - For each player j in [1, Nj]:
 - * If M_Coalitions(i, j) is equal to k:
 - Set N_ceros to 0.
 - For each player g in [1, Nj]:
 - + If M_Coalitions(i, g) > 0:
 - * Increment N_ceros by 1.
 - If N_ceros is equal to lg:
 - * Increment nMk by 1.
 - * Copy the coalition M_Coalitions(i, :) to Mk_Coalitions(nMk, :).
 - * Copy the coalition value V_Coalition(i) to Vk_Coalition(nMk).
 - * Break the loop for j.
 - iii. Initialize Sub_coal as a matrix of zeros with dimensions (nMk, Nj).
 - iv. For each coalition i in [1, nMk]:
 - Set j to 0.
 - For each player g in [1, Nj]:
 - * If Mk_Coalitions(i, g) > 0 and Mk_Coalitions(i, g) is not equal to k:
 - + Increment j by 1.
 - + Set Sub_coal(i, j) to Mk_Coalitions(i, g).
 - v. For each coalition i in [1, nMk]:
 - If lg is equal to 1:
 - * Set Vk_Sub_coal(i) to 0.
 - Else:
 - * For each coalition j in [1, n_coal]:
 - + If Sub_coal(i, :) is equal to M_Coalitions(j, :):
 - * Set Vk_Sub_coal(i) to V_Coalition(j).
 - vi. Set M_Shapley(k, lg) to the sum of (Vk_Coalition – Vk_Sub_coal).
 - vii. Clear Vk_Coalition, Vk_Sub_coal, Mk_Coalitions, and Sub_coal.
2. For each player S in [1, Nj]:
 - Set B(S) to (factorial(S-1) × factorial(Nj – S))/factorial(Nj).
3. For each player i in [1, Nj]:
 - Set Z to M_Shapley(i, :) element-wise multiplied by B.
 - Set Weight (i) to the sum of Z.

However, it is noteworthy that despite this limitation, the Shapley value contributes to a more equitable distribution of resources by allocating bandwidth proportionally according to the contribution of each node to the optimization process. This observation underscores the importance of considering not only the absolute number of resources allocated but also their equitable distribution among the nodes within the system.

Table 8. Shapley Matrix for the proposed scenario.

Player j	Contribution to the Coalition Containing j Players												$\varphi_j(v)$ [1×10^3]
	1	2	3	4	5	6	7	8	9	10	11	12	
1	0	10	4867	16,170	32,340	45,276	45,276	32,340	16,170	5390	1078	98	80.95
2	0	22	5419	18,150	36,300	50,820	50,820	36,300	18,150	6050	1210	110	90.88
3	0	82	6109	20,460	40,920	57,288	57,288	40,920	20,460	6820	1364	124	102.88
4	0	78	6060	20,295	40,590	56,826	56,826	40,590	20,295	6765	1353	123	102.02
5	0	86	6158	20,625	41,250	57,750	57,750	41,250	20,625	6875	1375	125	103.73
6	0	10	4867	16,170	32,340	45,276	45,276	32,340	16,170	5390	1078	98	80.95
7	0	270	7422	24,585	49,170	68,838	68,838	49,170	24,585	8195	1639	149	125.04
8	0	12	4959	16,500	33,000	46,200	46,200	33,000	16,500	5500	1100	100	82.60
9	0	22	5419	18,150	36,300	50,820	50,820	36,300	18,150	6050	1210	110	90.88
10	0	86	6158	20,625	41,250	57,750	57,750	41,250	20,625	6875	1375	125	103.73
11	0	86	6158	20,625	41,250	57,750	57,750	41,250	20,625	6875	1375	125	103.73
12	0	12	4959	16,500	33,000	46,200	46,200	33,000	16,500	5500	1100	100	82.60
$P(j)$	0.0833	0.0076	0.0015	0.0005	0.0003	0.0002	0.0002	0.0003	0.0005	0.0015	0.0076	0.0833	1150

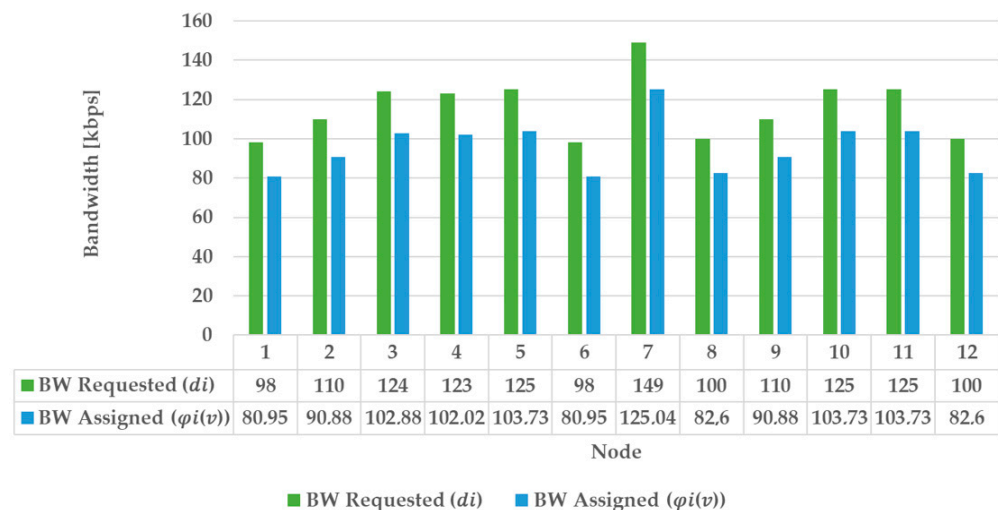


Figure 4. BW requested and BW assigned (Shapley) for a channel saturation state.

3.2. Resource Optimization of the Proposed Scenario Using Linear Programming Techniques

Linear programming has proven to be an invaluable tool in optimizing resources, such as bandwidth, by providing a systematic and efficient approach to the optimal allocation of these limited resources. Its mathematical methods enable precise modeling of the constraints and objectives of the problem, leading to optimal solutions that maximize system performance. In accordance with the proposed scenario, the linear programming problem can be formulated as follows:

$$\text{Maximize } \sum_{i=1}^n x_i \quad (12)$$

Subject to:

$$0 \leq x_i \leq d_i$$

$$\sum_{i=1}^n x_i \leq BW_T$$

where n , d_i , and x_i correspond to the number of nodes ($n = 12$), the bandwidth requested by node i , and the bandwidth assigned to node i , respectively. The LP problem was solved

using the Optimization Toolbox in MATLAB R2023a, employing various optimization methods. The objective function, constraints, and initial iteration point were organized in matrix form. The values were set as follows:

F : Vector of coefficients of the Objective function.

A, b : Inequality restrictions ($Ax \leq b$).

BW_T : 1150 Kbps.

lb, ub : Set the lower and upper limits allowed for each of the nodes, respectively.

Finally, the following expression is used to calculate the optimal solution to the problem using the “interiorpoint-legacy” algorithm, which yielded the best results compared to the “dual-simplex” and “interior-point” algorithms:

```
options = optimoptions("linprog","Algorithm","interior-point-legacy");
[x,fval] = linprog(F,A,b,[],[],lb,ub,options)
```

where x and $fval$ correspond to the solution vector and the maximum value that the objective function can reach. Figure 5 presents the results obtained for the proposed optimization model as a function of the bandwidth requested by each node.

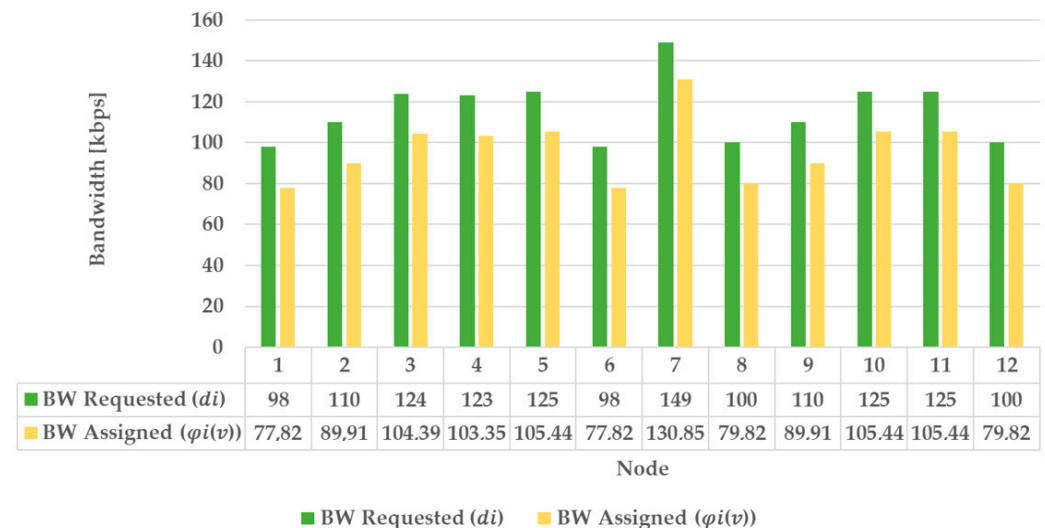


Figure 5. BW requested and BW assigned by Linear Programming for a channel saturation state.

The data presented in Figure 5 represent the results of the optimization process using linear programming to allocate bandwidth to each node in the system. It is observed that the amount of bandwidth requested by each node is generally greater than the amount allocated, once again indicating the presence of resource constraints in the system. However, upon comparing the assigned bandwidth values, it can be seen that in this case, the nodes have received, on average, a slightly larger amount of bandwidth in some nodes compared to the results obtained in Figure 4. This difference may be attributed to variations in optimization approaches between linear programming and the Shapley value method.

In terms of fairness in the distribution of resources, some variability is also observed among the nodes, suggesting that linear programming may prioritize the allocation of resources more uniformly among the nodes, aiming to maintain a balance in the system’s performance and satisfy the needs of the different nodes in a more proportional manner.

4. Discussion

The digital broadcasting system operates in two specific modes: hybrid mode and fully digital mode. In hybrid mode, simultaneous transmission of analog and digital components occurs, utilizing the channel assigned to the analog signal under the traditional

radio model. Conversely, in the fully digital mode, the transmitted signal utilizes the entire bandwidth allocated to the station, with its component being purely digital. It is noteworthy that digitalization aims to enhance service quality reaching the listener while optimizing transmission efficiency. Advantages of digital radio over analog radio include better audio quality, improved interference response, enhanced spectrum utilization, similar coverage with less power, and the incorporation of multimedia information such as graphics and video in the receiver, enabling services like displaying song lyrics, weather updates, and news.

In the context of digital radio, a significant challenge lies in optimizing bandwidth to meet each station's specific needs, given the limited spectrum and increasing service demand. Efficient allocation and utilization of this resource, alongside dynamic adaptation to demand fluctuations and radio-electric environment conditions, are crucial for optimal performance and effective bandwidth utilization. Dynamic optimization techniques like the Shapley value offer a promising approach by evaluating each station's relative contribution to bandwidth optimization, leading to more equitable and efficient resource allocations among stations, thereby enhancing resource management and user experience.

Using the Shapley value as a resource optimization strategy in an IBOC transmission scheme in FM presents several advantages. Firstly, it facilitates the fair and equitable evaluation of each node's relative contribution to the final outcome, particularly useful in systems with multiple agents like the 12 transmission stations in this case. Secondly, it aligns with fairness and justice concepts by assigning value to each node based on its marginal contribution to the system, reducing potential conflicts among nodes with different interests.

However, linear programming optimization also has advantages, offering precise and computationally efficient solutions when constraints and objectives are well defined. It allows explicit modeling of problem constraints, ensuring compliance with technical or regulatory requirements.

To assess the efficiency of the Shapley value in resource optimization compared to linear programming, a comparative statistical analysis was conducted for the proposed scenario. Table 9 presents the optimal bandwidth values for linear programming (BW_O), Shapley value (BW_{Sh}), as well as the differences (X_1 and X_2) between the bandwidth requested by each node (d_i) and the bandwidth assigned through linear programming and Shapley, respectively.

Table 9. Optimal BW-PL vs. BW-Shapley and differences in BW allocation according to channel state.

Node i	BW Requested d_i	Logic Channels BW [kbps]			
		BW_O	BW_{Sh}	X_1 $d_i - BW_O$	X_2 $d_i - BW_{Sh}$
1	98	77.8238	80.95	20.1762	17.05
2	110	89.9117	90.88	20.0883	19.12
3	124	104.3908	102.88	19.6092	21.12
4	123	103.3451	102.02	19.6549	20.98
5	125	105.4379	103.73	19.5621	21.27
6	98	77.8238	80.95	20.1762	17.05
7	149	130.8483	125.04	18.1517	23.96
8	100	79.8156	82.60	20.1844	17.4
9	110	89.9117	90.88	20.0883	19.12
10	125	105.4379	103.73	19.5621	21.27
11	125	105.4379	103.73	19.5621	21.27
12	100	79.8156	82.60	20.1844	17.4

The analysis conducted aims to determine which of the two optimization methods provided the most suitable adjustment to the requirements of each node, a criterion discernible through the technique resulting in the lowest disparity between the requested and allocated bandwidth for each node in an IBOC FM network. Accordingly, the following hypotheses are proposed [27]:

$$H_0 : \mu_x \leq \mu_y \rightarrow \mu_x - \mu_y \leq 0 \rightarrow \mu_z \leq 0 \quad (13)$$

$$H_a : \mu_x > \mu_y \rightarrow \mu_x - \mu_y > 0 \rightarrow \mu_z > 0$$

Here, μ_y and μ_x represent the means for the difference between requested bandwidth and assigned bandwidth through the PL optimization methods and the Shapley value, respectively. Hypothesis H_0 suggests a significant difference between means, with μ_x being lower than μ_y , indicating that the Shapley value leads to a more effective optimization process. Hypothesis H_a suggests the opposite. Additionally, a new variable Z is introduced to refine the hypotheses.

To evaluate these hypotheses, a paired t -test is employed, commonly used for assessing the statistical validity of differences between two random samples [28]. The process involves the following steps:

Step 1: Define a new random variable $Z = X - Y$ and calculate the mean (Z) and standard deviation (S_z) for Z . The resulting values are 0.0008 and 2.7544, respectively.

Step 2: Calculate the test statistic (d) using the expression:

$$d = \frac{\bar{Z}}{S_z} \sqrt{n} = \frac{0.0008}{2.7544} \sqrt{12} = 1.006 \times 10^{-3} \quad (14)$$

where d is the statistic value, and n is the number of samples for both proposed models.

Step 3: Establishing the acceptance range for H_0 as $\{t: t < T_{(\alpha, n-1)}\}$ at a 5% significance level ($\alpha = 0.05$) with $n - 1$ degrees of freedom; for this case, $T(0.05; 11) = 1.7959$. Defining the acceptance interval for H_0 as $(-\infty, 1.67959)$. Evaluation of the statistic d reveals that it falls within the acceptance interval, leading to the non-rejection of H_0 . Consequently, it can be concluded with 95% confidence that the Shapley value stands as an excellent alternative for executing high-quality optimization processes.

One potential disadvantage of the Shapley value is its computational cost, as calculating it may require evaluating all possible node coalitions, which can be complex in systems with a large number of agents. On the other hand, linear programming may be less flexible in situations where node relationships are complex or changing, as it requires a precise formulation of the problem and may not fully capture nonlinear interactions between nodes.

Both the Shapley value and linear programming offer valid approaches to resource optimization in IBOC transmission systems in FM, each with its own advantages and disadvantages. The choice between them will depend on the specific characteristics of the problem, such as the complexity of interactions between nodes, the availability of precise information, and computational efficiency requirements.

The use of the Shapley value can be considered a starting point of great importance for the optimization of resources in digital broadcasting over IBOC FM, through the use of techniques related to the theory of cooperative games. However, to advance in this field, it is necessary to evaluate in future work other cooperative techniques such as Nucleolus, Max–Min Fairness, and power indices, which can offer complementary perspectives and even the possibility of obtaining more equitable solutions in the allocation of digital broadcasting resources, considering that most of the studies that have been carried out by other researchers are mainly related to making better use of bandwidth through the development of advanced audio compression techniques and the use of more efficient modulation techniques. Additionally, the development of algorithms is proposed that reduce both the computational and temporal complexity of the techniques suggested for resource optimization, allowing their implementation in a more agile and efficient way in

practical environments. Finally, the use of genetic algorithms could become a promising strategy for resource optimization in IBOC FM digital broadcasting, offering a dynamic and adaptive approach, and with the possibility of significantly improving efficiency, latency, and quality.

5. Conclusions

In response to the need for equitable resource distribution among nodes in an RF network over IBOC FM, cooperative game theory with transferable utility is proposed as an optimization strategy. The goal is to maximize the allocated bandwidth for each node based on the service class, incorporating the Shapley value in a saturated channel state. This approach optimizes resource allocation by recognizing cooperation among nodes and ensuring adequate Quality of Service (QoS) levels. In the proposed scenario, the cooperative game is framed as a bankruptcy game, characterized by a ternary structure (N, d, C) . Here, $N = \{1, 2, \dots, n\}$ denotes the set of players, $d = \{d_1, d_2, \dots, d_n\}$ with $d_i \geq 0$ for all $i \in N$ represents the vector of demands, and C signifies the net value to be distributed among the elements of N . This formulation allows for the establishment of game imputations, leading to the estimation of the resulting payment vector using four proposed techniques. The outcomes of employing the Shapley value demonstrated outstanding results in equitable resource allocation, closely aligning with the specified service class requirements, with a confidence level of 95%. Furthermore, it was observed that the total sum of bandwidth assigned to each player aligns precisely with the total bandwidth available in the RF channel. This reinforces the effectiveness of the Shapley value as an optimal strategy for resource allocation in cooperative RF network scenarios.

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